

Derek Barbosa

Brockton, MA • (774) 381-1332 • derekbarbosa08@yahoo.com
github.com/derekb1999 • derekbarbosa.com

Education

University of Massachusetts Amherst

Bachelor of Science in Computer Science

Relevant Coursework: Databases, Web Development, Computer Networks

Amherst, MA

Expected May 2026

Massasoit Community College

Associate of Science in Computer Science (Transfer)

GPA: 3.6

Brockton, MA

May 2024

Technical Skills

Languages: Python, Java, JavaScript/TypeScript

Frameworks: FastAPI, React, Node.js, HTMX

Databases: PostgreSQL, Supabase

Tools: Git, Docker, Figma

Concepts: REST APIs, JWT Authentication, Relational Databases, Idempotency, Distributed Systems, Message Queues, Dead-Letter Queues (DLQs), Horizontal Scaling

Projects

Trailies - Full-Stack Trail Discovery and Review Platform

2025

- Built a full-stack trail discovery platform enabling authenticated users to browse trails and submit reviews
- Implemented authentication using JWTs stored in secure cookies with bcrypt password hashing
- Implemented role-based authorization with separate admin and user permissions
- Designed and implemented a RESTful API with 10+ endpoints using FastAPI and PostgreSQL to manage users, trails, and reviews
- Enabled real-time review submissions using HTMX, improving user interaction without full-page reloads
- Containerized the FastAPI backend and PostgreSQL database with Docker for reproducible local development

Distributed Event Ticketing Platform

2026

- Designed and built a microservices platform as part of an 8-person team
- Enforced purchase idempotency with unique keys to prevent duplicate purchases
- Owned the refund service end-to-end, enabling purchase refunds and automatic reserved seat release
- Implemented Redis-backed caching and asynchronous message queuing with dead-letter queues (DLQs) for fault-tolerant ticket purchasing
- Implemented horizontal scaling and reverse proxy load balancing with Docker and Caddy, using k6 to test system performance under concurrent load

Habitask - Semester Goal Planner and Tracker

2026

- Designed and built a semester goal planning and tracking platform using React, FastAPI, and Supabase as part of a 7-person development team
- Implemented Google OAuth 2.0 authentication for secure user sign-in and account creation
- Developed backend logic for friend request lifecycle management, including pending request tracking and duplicate request prevention
- Implemented dynamic friend status rendering based on user privacy settings and request state